

PDF: Prompt-guided Decoupled Feature Learning for Cloth-Changing Person Re-identification

Hao Chen, Gang Shen, Xuefei Lv, Ziwei Liu, Limin Su, Yunhai Tong, Shiliang Zhang

Abstract—Cloth-changing person re-identification (CC-ReID) aims to extract cloth-irrelevant visual features to match individuals across different outfits. While images contain rich visual cues, textual descriptions often provide explicit and structured attribute information, offering a potential pathway to decouple and suppress cloth-relevant features in visual representations. In this paper, we propose a novel prompt-guided decoupled feature learning framework that leverages text prompts to guide the learning of cloth-irrelevant features for CC-ReID. Given that human-annotated descriptions are typically unavailable in CC-ReID datasets, we explore the use of prompts generated by multimodal large language models (MLLMs) as a substitute. To extract cloth-relevant features, we design a prompt-guided visual feature extractor that learns attribute-aware features aligned with the prompts. A decoupled feature learning module is then employed to suppress the cloth-relevant components from the raw visual features. Through this prompt-guided learning strategy, our framework is able to construct cloth-irrelevant representations. Extensive experiments on multiple CC-ReID benchmarks demonstrate that our method outperforms existing state-of-the-art approaches, validating its effectiveness and robustness.

Index Terms—Cloth-changing person re-identification, vision-language models, prompt learning.

I. INTRODUCTION

PERSON re-identification (ReID) targets at matching a person of interest across non-overlapping cameras. Although significant improvement has been witnessed in conventional short-term person ReID [1], [2], it is still difficult to re-identify a person in the long-term scenario where people change their clothes either intentionally or unintentionally. Towards building an intelligent video monitoring system, cloth-changing person re-identification (CC-ReID) is hence proposed to alleviate the influence of changed clothes and improve the long-term person ReID performance.

CC-ReID is a challenging yet practical task that typically relies on extracting cloth-irrelevant person identity features, such as body shape and gait, to match a same person with changed clothes. Previous methods can be roughly categorized into single-modality [3], [4] and multi-modality [5],

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The source code is available at <https://github.com/chenhao2345/PDF>.

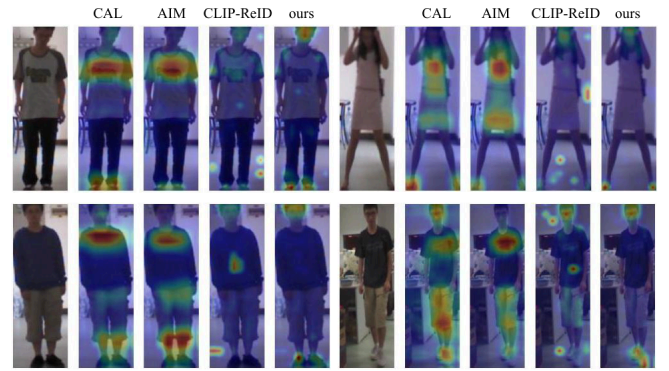


Fig. 1. Comparison of activation maps on the PRCC dataset. Compared to CAL [3] and AIM [7], which cover broad areas including clothing, CLIP-ReID [2] demonstrates enhanced discriminative power with more concentrated activations. Notably, our method achieves the highest precision by filtering out clothing-related noise and selectively highlighting only the cloth-irrelevant key regions.

[6] approaches. Single-modality approaches usually design dedicated modules or loss functions to extract desired cloth-irrelevant features from RGB images. Multi-modality approaches use extra information, such as human parsing mask or 3D body shape, to assist cloth-irrelevant feature extraction. However, the human parsing mask and 3D body shape can be easily affected by camera viewpoints and detection errors.

Textual descriptions of a person usually summarize human attribute information [8], [9], which has proven to be effective in person ReID. Compared with human parsing mask and 3D body shape, the attribute information summarized in the text descriptions can be more robust to detection box and viewpoint variance. As shown in Figure 1, previous CC-ReID methods usually include both cloth-relevant and cloth-irrelevant features. Compared with entangled visual features, textual attributes can be easier to be classified as cloth-relevant or cloth-irrelevant, which can help suppress undesired cloth-relevant features. Thus, we explore the possibility of using text prompts as guidance to learn purified cloth-irrelevant visual features for CC-ReID.

In this paper, we propose a Prompt-guided Decoupled Feature learning (PDF) framework for CC-ReID. Although human-annotated attribute labels can provide precise supervision for identity feature learning, such annotations are typically unavailable in existing CC-ReID datasets. To address this limitation, we explore replacing human-provided descriptions with automatically generated ones. Recent advances in multimodal large language models (MLLMs) [10], [11] have enabled accurate and detailed image captioning, making them a promising source of attribute-aware textual prompts for person images.

To fully exploit the potential of textual guidance, our PDF framework incorporates a cross-modal feature extractor and dedicated loss functions to align vision-language features and explicitly filter out cloth-related visual cues in the shared representation space.

Specifically, our PDF framework is built on a CLIP [12] network that contains an image encoder and a text encoder. Considering different images of the same person as positive samples, we first pull together positive embeddings from the image encoder as our baseline. On top of that, benefiting from the aligned vision-language feature space of CLIP, we design a cross-modal prompt-guided feature extractor that extracts visual features that correspond to cloth attributes in the text prompts. We further design a decoupled feature learning module to enhance the orthogonality between prompt-guided features and visual features. As the prompt-guided features are mainly associated to clothes, only cloth-irrelevant visual features are preserved in person representations. By combining the prompt-guided feature extractor and the decoupled feature learning module, our PDF framework optimally learn cloth-irrelevant identity features for CC-ReID.

In summary, our contributions are threefold:

- We investigate the effectiveness of the text modality in guiding visual feature learning. To the best of our knowledge, this is an initial attempt that employs MLLM-generated prompts as guidance to decouple cloth-irrelevant visual features for CC-ReID.
- We propose a prompt-guided decoupled feature learning framework for CC-ReID. With the MLLM-generated text prompts, our method combines a prompt-guided feature extractor and a decoupled feature learning module to learn cloth-irrelevant features.
- We conduct extensive experiments to validate the effectiveness of our proposed method. Our method outperforms previous CC-ReID methods on both single-domain discrimination ability and cross-domain generalization ability.

II. RELATED WORK

A. Person Re-identification

Conventional person ReID [1], [13], [14] typically assume that a person usually wears the same clothes, where the identity features are correlated to cloth attributes. However, such assumption is only valid for short-term person ReID. When it comes to the long-term scenario, the correlation between identity and clothes becomes an obstacle for cloth-changing person ReID.

To handle cloth variance, previous CC-ReID methods can be roughly categorized into two classes, *i.e.*, single-modality and multi-modality methods. Single-modality methods solely rely on the image modality to build identity representations. Yu *et al.* [15] propose cloth template based person ReID, which combines biometric features from a person image and cloth features from a cloth template image to match a target image with changed clothes. With both identity and cloth labels, Gu *et al.* [3] introduce a clothes-based adversarial loss (CAL) to mine cloth-irrelevant features from the original RGB

images by penalizing the predictive power of ReID models. On top of CAL, Yang *et al.* [7] propose a causality-based auto-intervention model (AIM) to mitigate cloth bias. Yu *et al.* [16] propose a GAN-based method to synthesize cloth-changing images. Han *et al.* [4] propose a clothing-change feature augmentation (CCFA) model to implicitly augment cloth-changing data. As extracting cloth-irrelevant features from the single image modality can be difficult, multi-modality methods extract features with the guidance of other modalities, such as pose [17], [18], 3D shape [6], silhouettes [5], [19], contour sketches [20] and radio signals [21]. Recently, Peng *et al.* [22] propose a masked attribute description embedding method that incorporates human attributes and a vision-language model to extract identity-robust features. Compared to conventional attribute recognition models which rely on pre-defined label sets and are sensitive to detection box accuracy, MLLM-generated textual descriptions provide more robust, semantically rich, and interpretable representations. They can seamlessly integrate multiple visual cues into natural language and generalize to unseen attributes.

B. Vision-language Models

Compared with widely-used ImageNet [23] pre-training that only consider the image modality, vision-language pre-training seeks to build the connection between vision and natural language. CLIP [12] and ALIGN [24] utilize a symmetric InfoNCE loss to train a pair of image and text encoders, which have shown impressive transferability and generalizability in downstream tasks. Luo *et al.* propose CLIP4Clip [25] that transfers the knowledge of the original CLIP model to video-language retrieval. CLIP4Cir [26] propose to merge image and text features to compose a multimodal query for composed image retrieval. DenseCLIP [27] converts the original image-text matching problem in CLIP to a pixel-text matching problem and uses the pixel-text score maps to guide the learning of dense prediction models. Instead of using fixed context words in text prompts, CoOp [28] proposes to model a prompt's context words with learnable vectors for image classification. CoCoOp [29] further learns a light-weight visual network to give meta tokens for each image, which is combined with a set of learnable context vectors.

Inspired by recent advances in vision-language models, researchers have attempted to apply CLIP to person ReID. Li *et al.* [2] propose a CLIP-ReID framework that incorporates ambiguous text descriptions with learnable tokens in a two-stage training strategy for person ReID. MP-ReID [30] leverages both learnable and large language model generated prompts to assist person ReID. For text-to-image ReID, Jiang *et al.* [31] propose a cross-modal Implicit Relation Reasoning and Aligning (IRRA) framework to learn local visual-textual token relations and enhance global image-text matching for text-to-image person ReID. Yan *et al.* [32] further perform fine-grained information excavation to mine cross-modal correspondences, while MANet [33] suppresses image-specific background noise through relation-guided localization. For cross-modal ReID, Chen *et al.* [34] introduce a learnable cluster-aware prompt for unsupervised visible-infrared (VI) ReID. CSDN [35] bridges the modality gap by

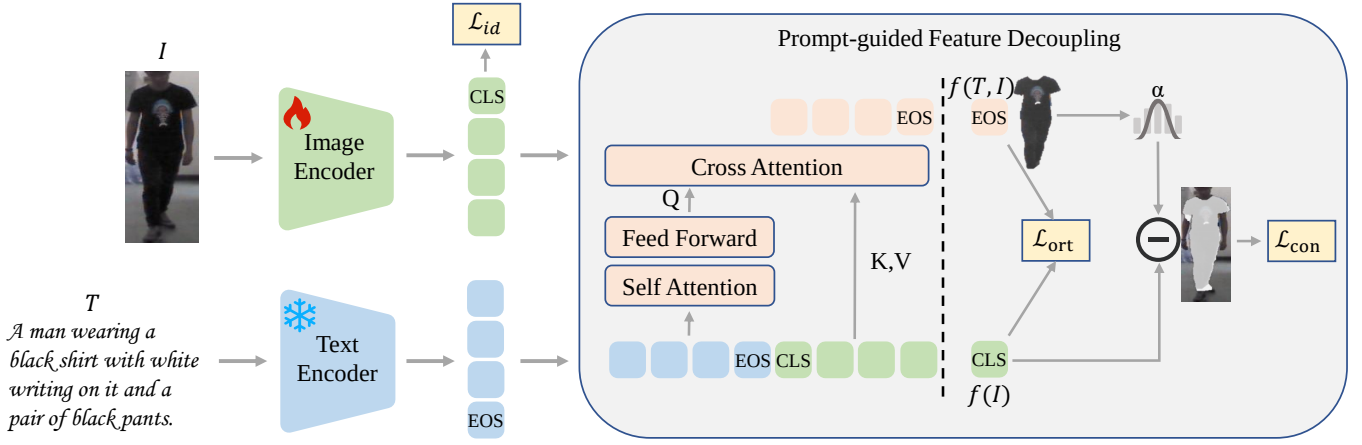


Fig. 2. Overview of our proposed PDF framework. Our proposed framework is composed of a visual encoder and a text encoder. During training, we only fine-tune the visual encoder, while the text encoder is frozen. Given a query image I and MLLM-generated text description T , we design three losses $\mathcal{L}_{id}$, $\mathcal{L}_{ort}$ and $\mathcal{L}_{flt}$ to decouple cloth-irrelevant features. After training, only the [CLS] token from the visual encoder is used for inference.

infusing visual features with high-level semantics, and Li *et al.* [36] generate modality-shared video-level language prompts to mitigate modality differences in video VI-ReID. In the video domain, Yu *et al.* [37] propose TF-CLIP to capture temporal information, whereas DS-VReID [38] utilizes dynamic prompts to construct 3D graphs for capturing spatial-temporal hierarchical relations. For domain adaptation, Li *et al.* [39] employ logical relation inference to increase robustness against camera style variations.

Regarding cloth-changing scenarios, Li *et al.* [40] design cloth-agnostic text prompts to guide the model in extracting identity features while attenuating clothes-related cues. Recently, Wei *et al.* [41] incorporate clothing masks with learnable prompts for CC-ReID, while Wang *et al.* [42] utilize prompts as transfer anchors for lifelong ReID across hybrid clothing states. Previous CLIP-based ReID methods typically rely on implicit learnable prompts which may still capture context information from clothes. Differently, our method filters out cloth-relevant information with MLLM-generated explicit prompts to explicitly decouple identity-invariant features, making it more suitable for CC-ReID.

III. METHODOLOGY

A. Overview

Given a query image I_q and a target image I_t of the same person, CC-ReID aims to maximize the representation similarity between I_q and I_t with different outfits. For query and target images, we use our visual encoder to extract visual identity representations $f(I_q)$ and $f(I_t)$. We optimize the network weights θ to obtain the optimal weights $\hat{\theta}$ that maximize the similarity between $f(I_q)$ and $f(I_t)$:

$$\hat{\theta} = \arg \max_{\theta} \cos(f(I_q), f(I_t)), \quad (1)$$

where $\cos(\cdot, \cdot)$ is the cosine similarity between two feature vectors.

In this paper, we introduce a Prompt-guided Decoupled Feature learning (PDF) framework for CC-ReID. Our PDF

framework leverages MLLM-generated text prompts to filter out cloth-relevant features and preserve cloth-irrelevant features in identity representations. Figure 2 summarizes our PDF framework that includes a visual encoder and a text encoder, which are initialized with CLIP pre-trained weights. We only train the visual encoder, while the text encoder is frozen during training. We first use a MLLM to generate cloth descriptions T for the query image I . We use the text feature $f(T)$ to extract the corresponding visual feature of the cloth $f_{cloth}(T, I)$. We proceed to design a feature augmentation module to enhance the feature decoupling. In the end, we combine an identity loss $\mathcal{L}_{id}$, an orthogonality loss $\mathcal{L}_{ort}$ and a filter loss $\mathcal{L}_{flt}$ to train our network:

$$\mathcal{L}_{total} = \mathcal{L}_{id} + \lambda_1 \mathcal{L}_{ort} + \lambda_2 \mathcal{L}_{flt}. \quad (2)$$

For inference, the generated prompts and the text encoder are discarded. Only the visual encoder is preserved to extract visual tokens as identity representations for CC-ReID.

B. Baseline

We build a simple yet effective baseline that enables the visual encoder to extract person identity representations. In our training phase, a mini-batch is composed of N_P identities, where each identity has N_K positive instances. Given a random image I belonging to the same class Y , we use the [CLS] token $f(I)$ from the visual encoder to calculate the identity loss. The identity loss is defined as a softmax cross-entropy loss:

$$\mathcal{L}_{id} = -\log \left(\frac{\exp(f_{CLS}(I_Y))}{\sum_{k=1}^K \exp(f_{CLS}(I_k))} \right), \quad (3)$$

where $Y \in \{1, 2, \dots, K\}$ is the ground truth labels, and $f_Y(I)$ is the prediction score of the class Y . A triplet loss [43] can

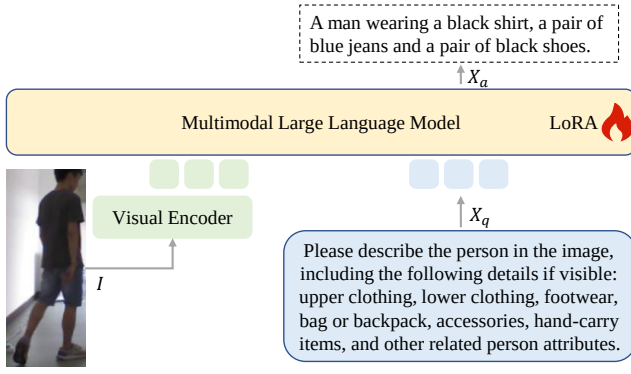


Fig. 3. Illustration of text description generation procedure.

be further added to enhance the discriminative ability on hard samples. The triplet loss is defined as:

$$\mathcal{L}_{tri} = \max(0, \|f_{CLS}(I) - f_{CLS}(I_p)\|_2^2 - \|f_{CLS}(I) - f_{CLS}(I_n)\|_2^2 + m), \quad (4)$$

where we randomly sample a positive image I_p and a negative image I_n for each image I , and m is a margin hyperparameter. By optimizing the identity loss, we aim to maximize the similarity between the positive image representations in the mini-batch, while minimizing the similarity between the negative image representations.

C. MLLM-based Prompt Generation

We leverage a multimodal large language model [10], [11] to generate descriptions for person images, which is illustrated in Figure 3. To keep our generated descriptions in a consistent format as previous text-to-image person ReID, we conduct an instruction tuning on CUHK-PEDES [44] and ICFG-PEDES [45] datasets. In the instruction tuning, each entry comprises a triplet (I, X_q, X_a) , where I is a person image, X_q is a unified textual instruction, and X_a is the ground-truth caption for the person. The X_q is defined as: “Please describe the person in the image, including the following details if visible: upper clothing, lower clothing, footwear, bag or backpack, accessories, hand-carry items, and other related person attributes”.

Given the visual tokens H_v and the instruction tokens H_q , the optimization objective of text generation is to maximize the probability of generating correct answer tokens H_a as follows:

$$\hat{\theta} = \arg \max_{\theta} p(H_a | H_v, H_q), \quad (5)$$

where θ contains the trainable parameters in the large language model θ_{LLM} , a trainable visual CLIP encoder θ_E and a projection layer θ_W . The visual encoder θ_E extracts patch tokens from the person image, followed by a trainable projection matrix θ_W to transform these patch tokens into visual tokens $H_v = \theta_W \cdot \theta_E(I)$. During offline training, we leverage LoRA [46] algorithm to efficiently fine-tune the textual description generation module. Using this module, we generate a textual description for each image.

D. Prompt-guided Feature Decoupling

Cloth-relevant and cloth-irrelevant features are usually entangled in visual embeddings. Simply training with the identity and triplet loss is prone to lead to sub-optimal performance. As text descriptions summarize explicit attribute information of a person image, we use MLLM-generated text descriptions to disentangle cloth-irrelevant features from the visual embeddings.

As shown in Figure 2, we design a cross-modal prompt-guided feature extractor to extract visual features that correspond to the text prompt. The inputs of a self-attention layer [47] contain a query (Q), a key (K) and a value (V). By calculating the similarity between the query Q and the key K, the self-attention layer learns to rearrange the value V. A Multi-Head self-Attention (MHA) can be defined as:

$$\text{MHA}(Q, K, V) = \text{softmax}\left(\frac{QK}{\sqrt{d}}\right)V, \quad (6)$$

where d is the embedding dimension.

The text embedding $f(T)$ is first processed by the self-attention layer and a feedforward layer, where the text token at the position [EOS] is served as Q, while the full text tokens are served as K and V:

$$f'_{EOS}(T) = \text{FFN}(\text{MHA}(f_{EOS}(T), f(T), f(T))). \quad (7)$$

Then, we send the visual and text tokens to a cross-attention layer, where the processed text token $f'(T)$ is served as Q, and the visual tokens $f(I)$ are served as K and V. In this way, our query text tokens first interact with other text tokens and then interact with visual tokens. The generated prompts are mainly related to person outfits, making the prompt-guided cloth feature $f_{cloth}(T, I)$ contain cloth-relevant visual features. The prompt-guided cloth feature $f_{cloth}(T, I)$ is defined as:

$$f_{cloth}(T, I) = \text{MHA}(f'_{EOS}(T), f(I), f(I)). \quad (8)$$

We design two losses to filter out cloth-relevant features from the image embedding. As the prompt-guided visual feature $f_{cloth}(T, I)$ mainly contains cloth-relevant features, we expect that the image embedding $f_{CLS}(I)$ can focus on the cloth-irrelevant identity features, which are orthogonal to $f_{cloth}(T, I)$. The orthogonality loss is defined as:

$$\mathcal{L}_{ort} = \mathbb{1}_{[c_i=c_j]} \mathbf{S}_{ij} + \mathbb{1}_{[c_i \neq c_j, y_i=y_j]} (1 - \mathbf{S}_{ij}), \quad (9)$$

where $\mathbf{S}_{ij} = \cos(f_{CLS}(I), f_{cloth}(T, I))$.

The orthogonality loss $\mathcal{L}_{ort}$ is composed of two terms. Here, $\mathbf{S}_{ij}$ denotes the cosine similarity between the visual feature $f_{CLS}(I)$ and the clothing feature $f_{cloth}(T, I)$. The first term penalizes the similarity $\mathbf{S}_{ij}$ between visual and clothing features of samples with the same clothing ($c_i = c_j$). The second term encourages the high similarity between samples belonging to the same identity ($y_i = y_j$) but wearing different clothes ($c_i \neq c_j$).

We design a filter loss to further decouple cloth-irrelevant features. The visual embedding $f_{CLS}(I)$ contains both cloth-relevant and irrelevant features, while the prompt-guided embedding $f_{cloth}(T, I)$ mainly contains cloth-relevant visual features. Thus, the cloth-irrelevant features can be defined as

the subtraction of $f_{CLS}(I)$ and $f_{cloth}(T, I)$. We add a random coefficient α on the cloth features to virtually augment the feature diversity. Here, we use a Gaussian distribution with a mean of 0.5 and a standard deviation of 0.5 to define an augmented cloth-irrelevant embedding:

$$f_{ir}(I) = f_{CLS}(I) - \alpha f_{cloth}(T, I), \text{ where } \alpha \sim \mathcal{N}(0.5, 0.5^2). \quad (10)$$

Then, we use a contrastive loss to maximize the similarity between the augmented cloth-irrelevant embedding $f_{ir}(I)$ and the other randomly augmented positive cloth-irrelevant embeddings $f_{ir}(I^+)$ of the same person, so that our model can be robust to cloth variance. The filter loss is defined as:

$$\mathcal{L}_{flt} = -\log \frac{\exp(\cos(f_{ir}(I), f_{ir}(I^+)) \cdot \tau)}{\sum_{j \in N} \exp(\cos(f_{ir}(I), f_{ir}(I_j)) \cdot \tau)}. \quad (11)$$

IV. EXPERIMENTS

A. Datasets and Evaluation Protocols

We evaluate our method on four widely-used PRCC, LTCC, VC-Clothes and LaST datasets. We report both Cumulative Matching Characteristics (CMC) Rank1 accuracy and mean Average Precision (mAP). Following previous methods [3], [48], three types of test settings are used in our experiments, including 1) general setting (both cloth-changing and cloth-consistent samples are used to calculate accuracy), 2) same-cloth setting (only cloth-consistent samples are used to calculate accuracy), and 3) cloth-changing setting (only cloth-changing samples are used to calculate accuracy).

The PRCC [20] dataset is collected from 3 cameras and each identity has clothes changing. For Camera C, the clothing of each person is different from the clothing they wore in Cameras A and B. There are 33,698 images from 221 identities in PRCC dataset.

The LTCC [17] dataset contains 17,138 images of 152 identities, capturing 478 different outfits from 12 camera views. On average, there are 5 different outfits for each cloth-changing identity in LTCC.

The VC-Clothes [49] dataset is a synthetic dataset by the game engine which contains 512 virtual identities of 19,060 images in 4 different scenes with remarkable clothes changes. There are 256 identities for training and the other unseen 256 identities for testing in VC-Clothes.

The LaST dataset [50] is a large-scale long-term person ReID dataset with larger spatial and temporal ranges, which exhibits rich variations in clothing, light, weather, and locations. It contains 5,000 identities and 71,248 images for training, 5,806 identities and 135,529 images for testing.

B. Implementation Details

All input images are resized to 384×128 . The image data augmentation contains random horizontal flipping and erasing. We use ViT-B/16 as our default visual encoder. Our PDF-large version uses ViT-L/14 visual encoder to fairly compare with MADE [22]. The text encoder is a CLIP text Transformer. An Adam optimizer with a weight decay rate of 5×10^{-4} is used to optimize our networks. The learning rate is set by default

to 3.5×10^{-7} with a linear warm-up scheme in the first 10 epochs. To accelerate convergence on synthetic data, we set the learning rate to 3.5×10^{-6} with a linear warm-up scheme in the first 10 epochs.. The total training epoch number is set to 50. The learning rate decays by a factor of 0.1 at the 20-th and 40-th epoch. The batch size is set to 64, where we randomly sample $N_P = 8$ persons and $N_K = 8$ images per person. We set $\lambda_1 = 0.5$ and $\lambda_2 = 1$ for balancing the three losses. We set the temperature $\tau = 16$ in the contrastive loss. The training of our framework takes about 2 hours on 1 NVIDIA 3090 GPUs.

C. Comparison with State-of-the-art Methods

We compare our proposed PDF method with state-of-the-art methods on both single-domain discrimination ability and cross-domain generalization ability to show the superiority of our method.

1) *PRCC and LTCC*: We compare our proposed method PDF with state-of-the-art methods on LTCC and PRCC datasets in Table I. Here, the compared methods include both single and multi-modality methods.

We first list the performance of representative general person ReID, including PCB [1], OSNet [13], HPM [14], IANet [51] and ISP [52]. These methods leverage local features or attention mechanism to enhance person identity representations. Meanwhile, based on author-released code, we also re-implement CLIP-ReID [2] that leverages a CLIP-pretrained backbone and learnable prompts to learn identity representations. However, CLIP-ReID shows inferior performance than state-of-the-art CC-ReID methods on PRCC and LTCC datasets. This is because CLIP-ReID designed for conventional short-term ReID, which neglects the cloth variance in prompt learning.

The single-modality CC-ReID methods solely consider RGB images for feature extraction, including CAL [3], AIM [7] and MCSC [53]. In addition to RGB images, the multi-modality CC-ReID methods use 3D mesh or text to assist visual feature extraction, including CLIP-ReID [2], 3DInvarReID [6], CLIP3DReID [48] and MIPL [41]. CLIP3DReID [48] distills both 3D mesh and text attributes into a unified network, showing strong cloth-changing performance. While MIPL [41] achieves impressive results on the PRCC dataset by leveraging auxiliary clothing masks, our PDF demonstrates superior generalizability on the LTCC General setting and the LaST dataset without requiring additional mask supervision. We further use a larger visual backbone ViT/L-14 to construct PDF-large, which outperforms the state-of-the-art attribute-based vision-language model MADE [22].

2) *LaST*: We proceed to compare our proposed method with state-of-the-art methods on LaST dataset in Table I. Our method PDF achieves a significant improvement over existing approaches, surpassing CAL [3] and the recent method proposed by Zhao et al. [54] across all reported metrics. Specifically, our approach PDF attains a mAP of 43.7% and Rank-1 accuracy of 78.1%, demonstrating its superior capability in a large-scale data scenario. In addition, our PDF-large significantly outperforms the state-of-the-art attribute-based vision-language model MADE [22].

TABLE I

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON LTCC AND PRCC DATASETS. “MODALITY” REFERS TO THE MODALITIES USED IN TRAINING. “SC” AND “CC” DENOTE SAME-CLOTH AND CLOTH-CHANGING SETTINGS. † INDICATES REPRODUCED RESULTS. BOLD NUMBERS DENOTE THE BEST RESULTS.

Method	Venue	Modality	PRCC				LTCC				LaST	
			SC		CC		General		CC		Rank1	mAP
			Rank1	mAP	Rank1	mAP	Rank1	mAP	Rank1	mAP		
PCB [1]	ECCV’18	RGB	99.8	97.0	41.8	38.7	65.1	30.6	23.5	10.0		
OSNet [13]	ICCV’19	RGB	-	-	-	-	67.9	32.1	23.9	10.8		
HPM [14]	AAAI’19	RGB	99.4	96.9	40.4	37.2	-	-	-	-		
IANet [51]	CVPR’19	RGB	99.4	98.3	46.3	46.9	63.7	31.0	25.0	12.6		
ISP [52]	ECCV’20	RGB	92.8	-	36.6	-	66.3	29.6	27.8	11.9		
CLIP-ReID [2]	AAAI’23	RGB+Text	100.0	99.5	57.0	55.3	73.0	36.5	38.3	15.9		
CAL [3]	CVPR’22	RGB	100.0	99.8	55.2	55.8	74.2	40.8	40.1	18.0	73.7	28.8
AIM [7]	CVPR’23	RGB	100.0	99.9	57.9	58.3	76.3	41.1	40.6	19.1	-	-
3DInvarReID [6]	ICCV’23	RGB+3D	-	-	56.5	57.2	-	-	40.9	18.9	-	-
MCSC [53]	TIP’24	RGB	99.8	99.8	57.8	57.3	73.9	40.2	42.2	19.4	-	-
CLIP3DReID [48]	CVPR’24	RGB+3D+Text	-	-	59.3	60.6	-	-	42.1	21.7	-	-
Zhao <i>et al.</i> [54]	TMM’24	RGB	99.7	99.8	65.8	57.3	-	-	43.4	18.2	73.2	29.8
MIPL [41]	TIP’25	RGB+mask+Text	-	-	71.0	67.0	75.1	38.8	-	-	-	-
PDF (ours)		RGB+Text	99.9	98.3	66.2	62.7	77.9	42.4	42.6	20.8	78.1	43.7
MADE [22]	TMM’25	RGB+Text	100.0	98.6	64.3	59.1	84.2	48.2	47.4	24.4	79.0	40.9
PDF-large (ours)		RGB+Text	100.0	98.7	68.7	64.2	81.1	47.7	50.5	24.9	83.3	52.5

TABLE II

RESULTS IN THE SAME-CLOTH AND CLOTH-CHANGING SETTINGS FOR THE DATA FROM A PART OF CAMERAS ON VC-CLOTHES.

Method	SC (cam2&3)		CC (cam3&4)	
	Rank1	mAP	Rank1	mAP
FSAM [19]	94.7	94.8	78.6	78.9
3DSL [55]	-	-	79.9	81.2
CAL [3]	95.1	95.3	81.4	81.7
Zhao <i>et al.</i> [54]	94.7	94.9	81.8	81.7
ours	94.3	94.0	83.1	82.5

TABLE III

RESULTS IN THE SAME-CLOTH AND CLOTH-CHANGING SETTINGS FOR THE DATA FROM ALL CAMERAS ON VC-CLOTHES.

Method	General (all cams)		SC (all cams)		CC (all cams)	
	Rank1	mAP	Rank1	mAP	Rank1	mAP
CAL [3]	92.9	87.2	96.0	95.7	85.8	79.8
ours	94.7	89.7	96.3	94.7	89.2	84.3

3) *VC-Clothes*: We compare our proposed method with state-of-the-art methods on VC-Clothes dataset. Following [3], we report results of same-cloth subset from camera 2&3 and cloth-changing subset from camera 3&4 in Table II. Our method achieves best performance in the cloth-changing setting, while our results in the same-cloth setting are slightly lower than CAL [3]. We also report results of same-cloth and cloth-changing data from all cameras in Table III. Our method achieves best performance in the general and cloth-changing settings.

TABLE IV

COMPARISON OF GENERALIZATION ABILITY ON LTCC→PRCC. THE SETTING REFERS TO TRAINING ON LTCC AND TESTING ON PRCC.

Method	SC		CC	
	Rank1	mAP	Rank1	mAP
CAL [3]	99.0	93.3	36.8	33.1
AIM [7]	99.6	96.0	40.6	38.6
PDF	99.7	94.8	50.2	44.1

TABLE V

COMPARISON OF GENERALIZATION ABILITY ON PRCC→LTCC. THE SETTING REFERS TO TRAINING ON PRCC AND TESTING ON LTCC.

Method	General		CC	
	Rank1	mAP	Rank1	mAP
CAL [3]	23.9	7.1	7.4	3.7
AIM [7]	22.1	6.3	6.1	3.0
PDF	66.1	29.8	34.4	15.7

4) *Cross-Domain Generalization Ability on PRCC and LTCC*: We further compare the cross-domain generalization ability between previous CC-ReID methods and our proposed PDF, as shown in Tables IV and V, respectively. The cross-domain evaluation denotes training a CC-ReID model on one dataset and directly testing it on another dataset. Here, based on the author-released code, we re-implement two CC-ReID methods, *i.e.*, CAL and AIM for the cross-domain evaluation.

In the LTCC→PRCC setting, our method PDF achieves the best performance with 50.2% Rank-1 and 44.1% mAP in the cloth-changing setting, outperforming both CAL [3] and AIM [7]. Similarly, in the PRCC→LTCC setting, PDF also shows a significant improvement, achieving 66.1% Rank-

TABLE VI

ABLATION STUDY OF TRAINING LOSSES ON LTCC DATASET. TRAINING ONLY WITH $\mathcal{L}_{id}$ REFERS TO OUR BASELINE. TRAINING WITH THE THREE LOSSES IN EQUATION (2) REFERS TO OUR FULL PDF METHOD.

$\mathcal{L}_{id}$	$\mathcal{L}_{tri}$	$\mathcal{L}_{flt}$	$\mathcal{L}_{ort}$	General		CC	
				R1	mAP	R1	mAP
✓				53.5	22.8	25.0	11.0
✓	✓			72.0	38.8	37.5	18.3
✓		✓		76.5	42.4	41.8	20.7
✓		✓	✓	77.9	42.4	42.6	20.8

1 and 29.8% mAP in the general setting, and 34.4% Rank-1 and 15.7% mAP in the cloth-changing setting. This is because CAL and AIM only rely on the image inputs, which can be easily influenced by domain gaps, such as illumination, view-point and camera property variance. In contrast, text inputs are correlated with concrete attribute descriptions, which are more robust to domain variance. These results clearly demonstrate the superior generalization capability of our approach when transferred across datasets with different data distributions and capture conditions.

D. Ablation Study

1) *Training Loss*: To decouple cloth-irrelevant features from visual embeddings, our proposed method PDF incorporates an identity loss $\mathcal{L}_{id}$, an orthogonality loss $\mathcal{L}_{ort}$ and a filter loss $\mathcal{L}_{flt}$ to train our visual encoder. To validate the effectiveness of each component, we conduct ablation experiments by gradually adding one of them onto the baseline. As shown in Table VI, when only training with $\mathcal{L}_{id}$, our baseline achieves sub-optimal results on LTCC dataset. When we add the orthogonality loss $\mathcal{L}_{ort}$ onto the baseline, our network achieves slightly better results on the cloth-changing setting. When we add the filter loss $\mathcal{L}_{flt}$ onto the baseline, our network achieves significantly better results on both general and cloth-changing settings. When both the orthogonality loss $\mathcal{L}_{ort}$ and the filter loss $\mathcal{L}_{flt}$ are added onto the baseline, our full PDF method achieves the optimal CC-ReID performance.

2) *Random Coefficient*: We add a random coefficient α on the cloth-relevant features to virtually augment the feature diversity in the filter loss, so that our identity representations can be more robust to cloth variance. We compare several random coefficient sampling methods in Table VII, including fixed values $[0, 0.5, 1]$, uniform distribution $\mathcal{U}$ and gauss distribution $\mathcal{N}$.

The parameter α essentially functions as a decoupling intensity regulator. Theoretically, different images contain varying degrees of clothing-related noise. A fixed α might lead to either insufficient noise suppression (when α is small) or the inadvertent removal of identity-consistent information (when inadvertent is too large). As shown in Table VII, when $\alpha = 0$, the cloth-irrelevant embedding $f_{ir}(I)$ fails to suppress cloth-relevant features, leading to unsatisfactory results on the cloth-changing setting. When $\alpha = 1$, although the performance in the CC setting is superior to $\alpha = 0$, it results in a significant performance drop in the General setting (75.1% Rank-1). This

TABLE VII

COMPARISON OF RANDOM COEFFICIENT SAMPLING METHODS ON LTCC DATASET.

Coefficient	General		CC	
	Rank1	mAP	Rank1	mAP
$\alpha = 0$	76.3	42.1	38.0	20.0
$\alpha = 0.5$	76.5	42.6	41.1	20.6
$\alpha = 1$	75.1	42.1	39.8	20.4
$\alpha \sim \mathcal{U}(0, 1)$	77.5	42.6	41.6	20.8
$\alpha \sim \mathcal{N}(0, 1^2)$	76.1	42.4	41.7	20.7
$\alpha \sim \mathcal{N}(0.5, 0.5^2)$	77.9	42.4	42.6	20.8

TABLE VIII

COMPARISON OF DIFFERENT FUSION STRUCTURES ON LTCC DATASET.

Structure	General		CC	
	Rank1	mAP	Rank1	mAP
1) Concat+S-Attn	76.1	40.5	38.5	19.4
2) C-Attn	75.3	42.9	42.1	20.9
3) ours	77.9	42.4	42.6	20.8

indicates that a deterministic full subtraction is overly aggressive and may lead to over-decoupling, where identity-related information semantically overlapping with clothing attributes is inadvertently suppressed. We observe that random sampling methods consistently outperform fixed values. This superiority stems from the fact that stochastic coefficients simulate a broader range of clothing interference levels during training. By sampling $\alpha \sim \mathcal{N}(0.5, 0.5^2)$, the model is encouraged to learn identity representations that remain stable across diverse decoupling intensities, effectively acting as a regularizer that prevents the network from over-fitting to specific clothing appearances. The results validate the effectiveness of our proposed feature augmentation method with the random coefficient.

3) *Fusion Structure*: To evaluate the effectiveness of our proposed text-visual feature fusion strategy, we compare different fusion structures on the LTCC dataset, as shown in Table VIII. 1) The "concat+self" variant, which concatenates features followed by a self-attention layer, achieves limited performance, especially under the CC protocol. 2) The "only cross" structure, which relies solely on a cross-attention layer, improves the performance notably in terms of mAP. 3) Our

TABLE IX

COMPARISON OF DIFFERENT DESCRIPTIONS ON LTCC DATASET. 'ATTRIBUTES' DENOTES ATTRIBUTES PREDICTED BY MADE [22]. 'SFT' DENOTES OUR PROPOSED SUPERVISED FINE-TUNING METHOD.

Description	General		CC	
	Rank1	mAP	Rank1	mAP
Attributes [22]	77.7	42.4	41.8	20.8
LLaVA (w/o SFT) [10]	77.9	42.4	41.8	20.8
QWen-VL (w/o SFT) [11]	77.5	42.4	41.6	20.7
LLaVA (w/ SFT) [10]	77.7	42.4	42.6	20.8
QWen-VL (w/ SFT) [11]	77.9	42.4	42.6	20.8

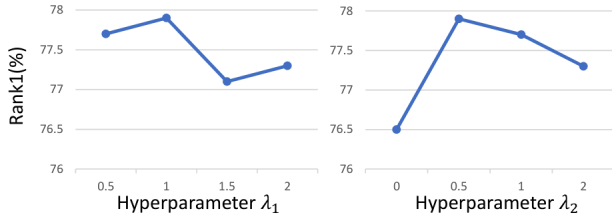


Fig. 4. Hyper-parameter tuning results of general setting on LTCC datasets.

proposed fusion structure achieves the best overall results. These results demonstrate the effectiveness of our design in leveraging complementary information from both modalities and highlight the importance of a well-balanced integration of self- and cross-attention mechanisms in multi-modal feature fusion.

4) *Choice of Description*: We investigate the impact of different description sources on our method by comparing performance when using traditional predicted attributes [22] and image-text descriptions generated by LLaVA [10] and QWen-VL [11], as shown in Table IX. Compared to traditional attributes that rely on a predefined set of labels, MLLM-generated descriptions provide richer semantic context. Notably, in the challenging Cloth-Changing (CC) setting, MLLM-based descriptions (w/ SFT) achieve a Rank-1 accuracy of 42.6%, outperforming the traditional attribute baseline (41.8%), which demonstrates the superiority of fine-grained textual descriptions in capturing identity-invariant features.

To further assess the sensitivity and robustness of our framework, we evaluate the impact of prompt quality and content variations from two perspectives. First, we examine sensitivity to prompt quality by comparing MLLMs with and without supervised fine-tuning (SFT). As shown in Table IX, while "off-the-shelf" MLLMs (w/o SFT) provide a solid baseline (e.g., 41.8% Rank-1 for LLaVA), the task-specific refinement through SFT further elevates performance to 42.6% in the CC setting. This indicates that while the framework is effective with generic descriptions, it can successfully leverage higher-quality, identity-focused prompts to solve cloth-changing challenges. Second, we evaluate robustness to content and stylistic changes by employing two distinct MLLM backbones. The results indicate that our approach exhibits consistent performance across both LLMs, achieving comparable results under the General and CC protocols. Specifically, QWen-VL slightly outperforms LLaVA with a Rank-1 accuracy of 77.9% vs. 77.7% in the General setting. The negligible performance gap (within 0.2% Rank-1) despite the differing linguistic styles and vocabularies of LLaVA and QWen-VL confirms that our method is highly robust to variations in description content, demonstrating low sensitivity to the specific choice of MLLM backbone.

5) *Hyper-parameter Analysis*: To investigate the impact of hyper-parameter choices on model performance, we conduct a sensitivity analysis on the LTCC dataset under the general setting. As shown in Figure 4, we vary the values of λ_1 and λ_2 of Equation (2) within a reasonable range. The performance,

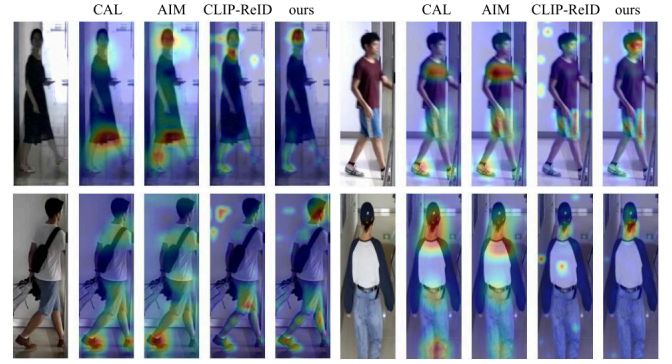


Fig. 5. Comparison of activation maps on the LTCC dataset. Compared to CAL [3], AIM [7] and CLIP-ReID [2], our method achieves the highest precision by filtering out clothing-related noise and selectively highlighting only the cloth-irrelevant key regions.

measured by Rank-1 accuracy, demonstrates that the model is relatively robust to the changes in both parameters. For λ_1 , the best performance is observed when $\lambda_1 = 1.0$, while for λ_2 , the optimal value is 0.5. These results suggest that a balanced regularization strength is beneficial for achieving better performance.

E. Qualitative Results

1) *Activation Maps*: To further demonstrate the superiority of our approach, we compare the activation maps of our method with the state-of-the-art CLIP-ReID. As illustrated in Figure 5, while CLIP-ReID [2] exhibits enhanced discriminative power with more concentrated activations than earlier baselines [3], [7], it still inevitably attends to clothing-related areas. Notably, our method achieves the highest precision by filtering out clothing-related noise and selectively highlighting only the cloth-irrelevant key regions. This confirms that the fine-grained semantic descriptions from MLLMs provide more explicit guidance for the model to focus on identity-invariant cues, which is essential for overcoming clothing variations.

2) *Retrieval Results*: To further validate the effectiveness of our proposed modules, we compare top-5 retrieved results between our baseline method and our full method in Figure 6. Even with the strong CLIP weights, we can observe that our baseline is still prone to fail when two different persons wear similar clothes. Our baseline preferentially picks images with similar clothes in the gallery set as positives. For the first example, our baseline is influenced by the similar white shirt, black jacket and jeans. For the second example, our baseline is influenced by the similar dark t-shirt and dark shorts. In contrast, our proposed PDF method is more robust to similar clothes. This is because we use our prompt-guided feature extractor and decoupled feature learning to filter out cloth-relevant features. The retrieval results confirm the strong ability of our proposed method in learning purified cloth-irrelevant features for CC-ReID.

F. Complexity Analysis

In the training phase, our proposed method PDF leverages MLLM-generated textual descriptions to filter out cloth-



Fig. 6. Comparison of top-5 retrieved results between the baseline and our full PDF method. Examples are randomly sampled from PRCC dataset. Green boxes denote correct results. Red boxes denote false results. Our baseline is prone to fail when facing similar clothes, while our full PDF method filters out cloth features, which is hence more robust to similar clothes.

TABLE X
COMPARISON OF FEATURE DIMENSION AND TOTAL INFERENCE TIME (IN SECONDS) ON THE PRCC AND LAST TEST SETS. ALL MODELS ARE EVALUATED USING THE SAME BATCH SIZE AND ONE 3090 GPU.

	Dim(f)	PRCC	LaST
CAL [3]	4096	50.78s	716.78s
PDF (ours)	512	50.67s	657.37s

relevant features from visual embeddings. The textual description generation with MLLM can be conducted offline, which thus has a limited influence on the training speed. In the inference phase, we only use the visual encoder to extract person representations, while the generated prompts and the text encoder are discarded. On the other hand, state-of-the-art methods, such as CAL [3] and AIM [7], usually combine average and max pooling to extract representations for image matching. The dimension of representations for CAL and AIM is 4096, while the dimension of our method is 512 for the ViT/B-16 backbone, making our PDF method faster in the representation matching phase. As shown in Table X, our inference speed is comparable with CAL on the middle-scale PRCC and outperforms CAL on the large-scale LaST. To conclude, the usage of the multimodal large language model and the CLIP backbone has quite limited influence on our inference speed.

V. CONCLUSION

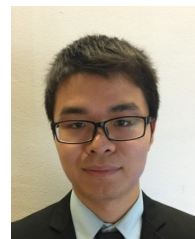
In this paper, we propose a novel prompt-guided decoupled learning framework for cloth-changing person re-identification. As text descriptions summarize concrete person attributes, we explore using multimodal large language model generated descriptions to learn purified cloth-irrelevant features. Specifically, we design a prompt-guided feature extractor to seek visual features that correspond to cloth attributes in the text prompts. We further design a decoupled feature learning module to filter out cloth-relevant features from raw image features. By combining the prompt-guided feature extractor and the decoupled feature learning, our method builds identity representations with decoupled cloth-irrelevant features for cloth-changing person re-identification. Experiments on both single-domain and cross-domain evaluation validate the effectiveness of our proposed method.

Despite the promising results, our method has certain limitations. The reliance on MLLM for prompt generation introduces additional computational overhead during the pre-processing stage, and the performance may be sensitive to the quality of generated descriptions in extremely low-resolution or cluttered environments. In the future, we will explore lightweight MLLM architectures for more efficient prompt generation and investigate adaptive semantic filtering mechanisms to further improve the robustness of identity representations.

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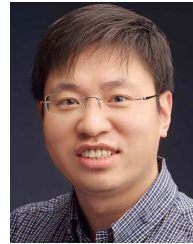
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